

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

module and configure it in 1:1 or 1:N mode for identification. Pins TX and RX of R305 sensor are connected to Arduino digital pins 2 and 3, which are used for serial communication.

The LCD display (LCD1) is used to display messages during action. Here, a 16x2 display is used; each character is made of 5x7 dot-matrix. Pins 3, 4, 5 and 6 of the LCD are the control lines connected to preset (PR1) output, pin 12 (Arduino), GND and pin 11 (Arduino). Pins 11, 12, 13 and 14 are data pins of the LCD that are connected to pins 7, 6, 5 and 4 of Arduino, respectively. Preset PR1 is used to adjust the contrast of the LCD display.

An electronic door-lock solenoid (connected across connector CON3) is basically an electromagnet made of a big coil of copper wire with an armature (slug of metal) in the middle. When the coil is energised, the slug is pulled into the centre of the coil. This allows the solenoid to move to one end.

The solenoid lock requires more current than what Arduino can provide. Therefore to operate the lock, a 5V relay (RL1) is used. CON3 is connected between normally open (N/O) contacts of RL1 and GND. The sequence of messages on the LCD from author's prototype are shown in Figs 2, 3 and 4.

Software

Programs named enroll and fingerprint use different functions like `getFingerprintEnroll(int,id)`, `Adafruit_Fingerprint(&mySerial)` and `getFingerprintEnroll(id)`. These functions are defined inside the library and pass arguments when called.

After uploading enroll in the Arduino, open serial monitor from Arduino IDE from Tools→Serial monitor options. Change baud rate below

DO-IT-YOURSELF

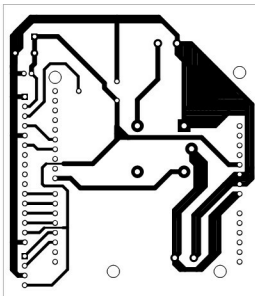


Fig. 5: Actual-size PCB pattern of the fingerprint door unlock system

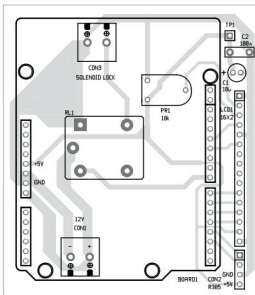


Fig. 6: Component layout of the PCB

the serial monitor window to 38400. Choose Newline option from the same place. Then, follow the instructions on the serial monitor. Place the finger on the fingerprint module. Type any whole number as the ID number. Press Send tab to send the ID number from the serial monitor to Arduino. This fingerprint gets converted into digital data and gets stored inside R305 module database.

More than 200 fingerprints can be stored on this system. Make sure

that each fingerprint has a unique ID number. This ID number will be used in the next program to identify the authenticated person's name. The serial monitor will guide the user as to when he or she should place the finger and when to remove it.

For debugging without an LCD display, make the same settings for the serial monitor after uploading Fingerprint program. This is used to compare the fingerprint in the sensor with stored prints. The serial monitor guides here also. The fingerprint program should be edited to change the name and ID numbers according to how users want.

Construction and testing

An actual-size, single-side PCB for the fingerprint door unlock system is shown in Fig. 5 and its component layout in Fig. 6. For convenience, we have designed the PCB as an Arduino shield. The users can modify the design as per requirement. Also, they can test PCB with Arduino board using a cable connector.

Make sure the baud rate given in the program is correct. Baud rate for the serial monitor can be anything but the baud rate for R305 sensor should match that given in its datasheet. Baud rate may vary with different versions of the sensor. It is given in the program like `Serial.begin(38400)` [baud rate for serial monitor]; `fingerprint.begin(57600)` [baud rate for sensor]. Reset Arduino board before validation of the fingerprint. **EFY**

